Speech Emotion Recognition System Using Python

Internship Project Report

Project Title: Speech Emotion Recognition System Using Python

Internship: Pinnacle Grid Internship

Task: Task-3 – Speech Emotion Recognition System

Submitted By: Tanishka Jain

Technology Stack: Python, Librosa, Scikit-learn, Streamlit, NumPy, Pandas

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1. Abstract

Speech Emotion Recognition (SER) is one of the most important applications of Artificial Intelligence and Machine Learning. The primary objective of this project is to develop a machine learning model capable of identifying human emotions from speech recordings. Human speech contains valuable emotional information that can be extracted using digital signal processing techniques and machine learning algorithms.

In this project, the RAVDESS Emotional Speech Audio Dataset was used to train a Random Forest Classifier. Audio features were extracted using Mel Frequency Cepstral Coefficients (MFCC), which are among the most effective features for speech analysis. The extracted features were used to train the model, which was then tested on unseen audio samples.

A Streamlit-based web application was also developed to provide a simple graphical interface where users can upload a WAV audio file and receive the predicted emotion. The trained model achieved an accuracy of approximately **61.11%**, demonstrating the capability of machine learning in recognizing emotions from speech.

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2. Introduction

Human communication is not limited to spoken words; emotions expressed through voice play a significant role in conveying meaning. Speech Emotion Recognition aims to identify emotional states such as happiness, sadness, anger, fear, calmness, disgust, surprise, and neutrality from speech signals.

With advancements in Artificial Intelligence, computers can now analyze speech patterns to determine emotional states. This technology has numerous applications, including virtual assistants, customer support systems, healthcare, education, entertainment, and security.

The goal of this project was to create a speech emotion recognition system using Python and machine learning techniques that can classify emotions from audio recordings.

3. Problem Statement

Human emotions are difficult for machines to understand directly. Traditional software processes textual or numerical data efficiently but struggles with emotional understanding. Manual emotion analysis is time-consuming and subjective.

Therefore, there is a need for an automated system capable of recognizing emotions from speech using machine learning techniques.

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4. Objectives

The objectives of this project are:

- To build a Speech Emotion Recognition System using Python.
- To extract meaningful audio features using MFCC.
- To train a machine learning model using the extracted features.
- To classify emotions from speech recordings.
- To evaluate model performance using testing data.
- To develop a user-friendly web application using Streamlit.
- To upload the project to GitHub and document the implementation.

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5. Scope of the Project

This project has applications in:

- Virtual assistants
- Call center analytics
- Mental health monitoring
- Online education
- Human-computer interaction
- Customer feedback analysis
- Smart healthcare systems

- Security systems

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6. Software and Hardware Requirements

Software

- Python 3.12
- Visual Studio Code
- Streamlit
- GitHub
- Git

Python Libraries

- NumPy
- Pandas
- Librosa
- Joblib
- Scikit-learn
- Matplotlib

Hardware

- Windows 10/11
- Intel Processor
- 8 GB RAM
- Internet Connection

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7. Technologies Used

Python

Used as the primary programming language.

Librosa

Used for audio processing and feature extraction.

NumPy

Used for numerical computations.

Scikit-learn

Used to train and evaluate the Random Forest model.

Joblib

Used to save and load the trained machine learning model.

Streamlit

Used to build a simple web interface for prediction.

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8. Dataset Description

The project uses the **RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song)** dataset.

The dataset contains speech recordings from multiple actors expressing different emotions.

The emotions included are:

- Neutral
- Calm
- Happy
- Sad
- Angry
- Fearful
- Disgust
- Surprised

The dataset consists of **1440 audio files**, each stored in actor-specific folders.

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9. Methodology

The project follows these steps:

1. Collect the dataset.
2. Read audio files.
3. Extract MFCC features.
4. Prepare training data.
5. Split dataset into training and testing sets.
6. Train Random Forest Classifier.
7. Evaluate the model.
8. Save the trained model.
9. Build Streamlit application.
10. Predict emotions from uploaded audio.

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10. Feature Extraction

The most important step is feature extraction.

The project uses **Mel Frequency Cepstral Coefficients (MFCC)** because they effectively represent speech characteristics.

Steps include:

- Load WAV file
- Convert to numerical waveform
- Generate MFCC coefficients
- Calculate mean values
- Use extracted features as model input

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11. Machine Learning Model

The Random Forest Classifier was selected because it:

- Handles multi-class classification

- Reduces overfitting
- Works well with numerical features
- Provides stable performance

The model was trained using the extracted MFCC features.

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12. Model Training

The extracted features were divided into training and testing datasets using an 80:20 split.

The Random Forest model was trained using:

- Training features
- Emotion labels

After training, predictions were generated for the testing dataset.

The trained model was saved using Joblib for future use.

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13. Model Evaluation

The model achieved:

Accuracy: 61.11%

Evaluation was performed using:

- Accuracy Score

The prediction system successfully classified uploaded audio files into one of the predefined emotion categories.

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14. Streamlit Web Application

A Streamlit application was developed to make the system interactive.

Features include:

- Upload WAV file
- Audio playback

- Automatic feature extraction
- Emotion prediction
- User-friendly interface

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15. Project Workflow

1. User uploads speech file.
2. Audio is processed.
3. MFCC features are extracted.
4. Trained model predicts emotion.
5. Result is displayed on the web application.

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16. Results

The developed system successfully recognized emotions from speech recordings.

Key achievements:

- Successfully loaded 1440 audio files.
- Extracted MFCC features.
- Trained Random Forest model.
- Achieved **61.11% accuracy**.
- Developed a working Streamlit application.
- Successfully predicted emotions from uploaded WAV files.

(Insert your screenshots here: project folder, training output, prediction output, and Streamlit interface.)

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17. Advantages

- Easy to use
- Fast prediction

- Supports multiple emotions
- User-friendly interface
- Low computational cost
- Easily extendable
- Suitable for educational purposes

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18. Limitations

- Accuracy can be improved.
- Works only with WAV files.
- Performance depends on audio quality.
- Trained on a limited dataset.
- Real-time microphone input is not included.

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19. Future Scope

Future improvements include:

- Deep Learning models (CNN, LSTM)
- Real-time emotion detection
- Microphone recording support
- Multi-language emotion recognition
- Higher accuracy using advanced features
- Deployment on cloud platforms
- Mobile application development

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20. Conclusion

This project successfully demonstrates the implementation of a Speech Emotion Recognition System using Python and Machine Learning. The RAVDESS dataset was used to train a Random Forest Classifier after extracting MFCC features from speech recordings. The trained model achieved an accuracy of **61.11%** and successfully predicted emotions from unseen audio files. A Streamlit web application was developed to make the system interactive and user-friendly.

The project strengthened practical knowledge of Python, audio signal processing, machine learning model development, feature extraction, model evaluation, and web application deployment. It also highlights the growing importance of AI-powered speech analysis in healthcare, customer service, education, and intelligent virtual assistants.

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21. References

1. RAVDESS Emotional Speech Dataset.
2. Python Documentation.
3. Scikit-learn Documentation.
4. Librosa Documentation.
5. Streamlit Documentation.
6. NumPy Documentation.
7. Joblib Documentation.